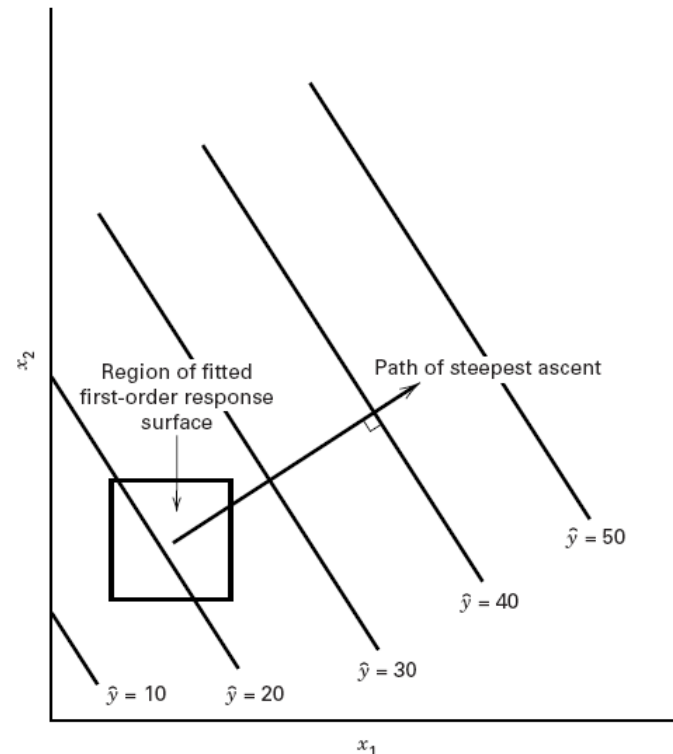


The Method of Steepest Ascent

- A procedure for moving sequentially from an initial “guess” towards region of the optimum
- Based on the fitted first-order model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

- Steepest ascent (or descent) is a gradient procedure



$$\hat{y} = 40.44 + 0.775x_1 + 0.325x_2$$

- To move away from the design center—the point ($x_1 = 0, x_2 = 0$)—along the path of steepest ascent, we would move 0.775 units in the x_1 direction for every 0.325 units in the x_2 direction.
- Thus, the path of steepest ascent passes through the point ($x_1 = 0, x_2 = 0$) and has a slope $0.325/0.775$.
- Decide a step size: $\Delta x_1 = 1$ (that is, 5 minutes of reaction time)
- Therefore, the steps along the path of steepest ascent are $\Delta x_1 = 1$ and $\Delta x_2 = (0.325/0.775) = 0.42$.

General Algorithm for Determining The Path

Assume that the point

$$x_1 = x_2 = \cdots = x_k = 0$$

is the base or origin point.

1. Choose a step size in one of the process variables, say Δx_j . Usually, we would select the variable we know the most about, or we would select the variable that has the largest absolute regression coefficient $|\hat{\beta}_j|$.
2. The step size in the other variables is

$$\Delta x_i = \frac{\hat{\beta}_i}{\hat{\beta}_j / \Delta x_j} \quad i = 1, 2, \dots, k \quad i \neq j$$

3. Convert the Δx_i from coded variables to the natural variables.

■ **TABLE 11.3**

Steepest Ascent Experiment for Example 11.1

Steps	Coded Variables		Natural Variables		Response y
	x_1	x_2	ξ_1	ξ_2	
Origin	0	0	35	155	
Δ	1.00	0.42	5	2	
Origin + Δ	1.00	0.42	40	157	41.0
Origin + 2Δ	2.00	0.84	45	159	42.9
Origin + 3Δ	3.00	1.26	50	161	47.1
Origin + 4Δ	4.00	1.68	55	163	49.7
Origin + 5Δ	5.00	2.10	60	165	53.8
Origin + 6Δ	6.00	2.52	65	167	59.9
Origin + 7Δ	7.00	2.94	70	169	65.0
Origin + 8Δ	8.00	3.36	75	171	70.4
Origin + 9Δ	9.00	3.78	80	173	77.6
Origin + 10Δ	10.00	4.20	85	175	80.3

TABLE 11.4**Data for Second First-Order Model**

Natural Variables		Coded Variables		Response
ξ_1	ξ_2	x_1	x_2	y
80	170	-1	-1	76.5
80	180	-1	1	77.0
90	170	1	-1	78.0
90	180	1	1	79.5
85	175	0	0	79.9
85	175	0	0	80.3
85	175	0	0	80.0
85	175	0	0	79.7
85	175	0	0	79.8

$$\hat{y} = 78.97 + 1.00x_1 + 0.50x_2$$

TABLE 11.5

Analysis of Variance for the Second First-Order Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P-Value
Regression	5.00	2			
Residual	11.1200	6			
(Interaction)	(0.2500)	1	0.2500	4.72	0.0955
(Pure quadratic)	(10.6580)	1	10.6580	201.09	0.0001
(Pure error)	(0.2120)	4	0.0530		
Total	16.1200	8			

The interaction and pure quadratic checks imply that the first-order model is not an adequate approximation. This curvature in the true surface may indicate that we are near the optimum. At this point, additional analysis must be done to locate the optimum more precisely.

Second-Order Models in RSM

When the curvature is detected, the experimenter is relatively close to the optimum, so a model that incorporates curvature is usually required to approximate the response.

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{ii} x_i^2 + \sum_{i < j} \beta_{ij} x_i x_j + \varepsilon$$

- Second-order models are used widely in practice
- The Taylor series analogy
- Optimization is easy
- There is a lot of empirical evidence that they work very well

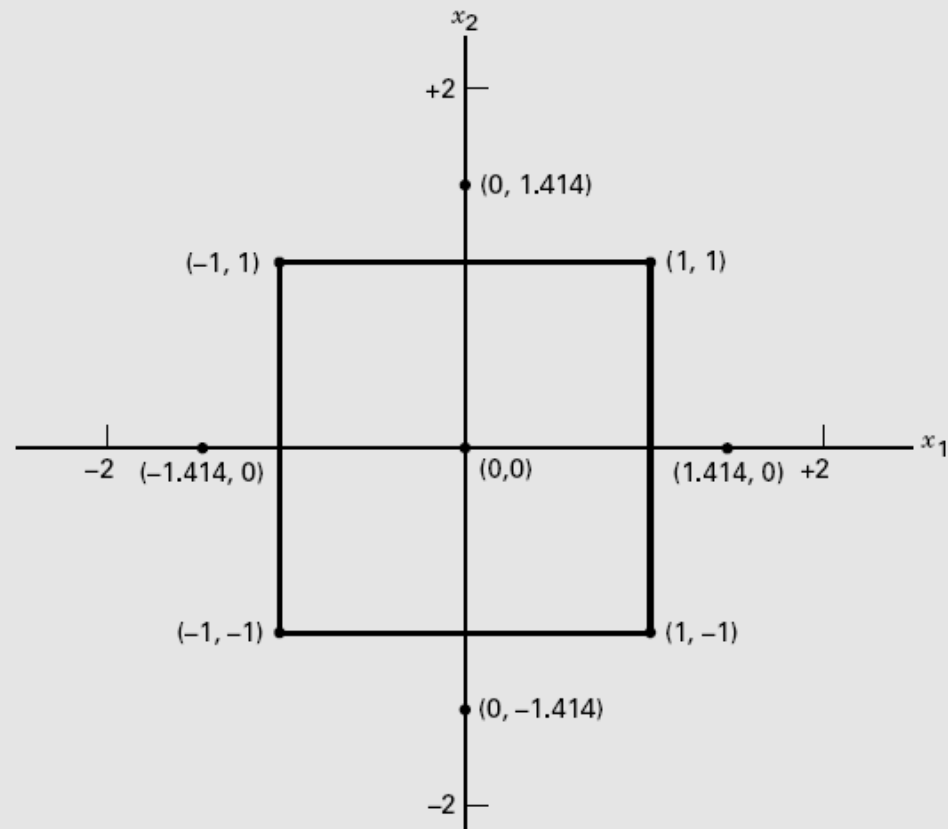
- We need more data
- This design is called a **central composite design (CCD)**
- Rotatable CCD
- Fit a second-order model with the data

TABLE 11.6

Central Composite Design for Example 11.2

Natural Variables		Coded Variables		
ξ_1	ξ_2	x_1	x_2	y_1 (Yield)
80	170	-1	-1	76.5
80	180	-1	1	77.0
90	170	1	-1	78.0
90	180	1	1	79.5
85	175	0	0	79.9
85	175	0	0	80.3
85	175	0	0	80.0
85	175	0	0	79.7
85	175	0	0	79.8
92.07	175	1.414	0	78.4
77.93	175	-1.414	0	75.6
85	182.07	0	1.414	78.5
85	167.93	0	-1.414	77.0

$$\hat{y} = 79.94 + 0.995x_1 + 0.515x_2 - 1.376x_1^2 - 1.001x_2^2 + 0.25x_1x_2$$



■ FIGURE 11.10 Central composite design for Example 11.2

Central Composite Design (CCD)

- Popular class of designs used for fitting second-order models.
- Generally, the CCD consists of
 1. a 2^k factorial (or fractional factorial) design with n_F factorial runs,
 2. $2k$ axial or star runs,
 3. n_C center runs.

Determine The α in The Axial Runs

- Principle I: set $\alpha = \sqrt{k}$, put all the factorial and axial design points on the surface of a sphere of radius k .
- Principle II: for the prediction $\hat{y}$ from the second-order model

$$y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \beta_{ii} x_i^2 + \sum_{i < j} \sum \beta_{ij} x_i x_j + \varepsilon$$

$V[\hat{y}(x)]$ is the same at all points x : $\alpha = (n_F)^{1/4}$

- A central composite design is made rotatable by the choice of α .
- Generally, three to five center runs are recommended.

(1) Rotatable CCD $\alpha = 8^{1/4} \approx 1.682$

Run	x_1	x_2	x_3
1	1	1	1
2	1	1	-1
3	1	-1	1
4	1	-1	-1
5	-1	1	1
6	-1	1	-1
7	-1	-1	1
8	-1	-1	-1
9	1.682	0	0
10	-1.682	0	0
11	0	1.682	0
12	0	-1.682	0
13	0	0	1.682
14	0	0	-1.682
15	0	0	0

(2) Equal-distance (spherical) CCD $\alpha = \sqrt{3} \approx 1.732$

Run	x_1	x_2	x_3
1	1	1	1
2	1	1	-1
3	1	-1	1
4	1	-1	-1
5	-1	1	1
6	-1	1	-1
7	-1	-1	1
8	-1	-1	-1
9	1.732	0	0
10	-1.732	0	0
11	0	1.732	0
12	0	-1.732	0
13	0	0	1.732
14	0	0	-1.732
15	0	0	0